Triage-Bot: An Assistive Triage Framework

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***Abstract*—Prompt delivery of healthcare services is critical for saving lives. Streamlining the collection of patients' health data through automation can minimize waiting times in emergency departments and enhance access to evidence-based care. We propose an innovative solution, Triage-Bot, which leverages the latest advancements in Artificial Intelligence to elevate the standard of patient care. Triage-Bot is a software as-a-service that allows patients or their caregivers to authenticate themselves by presenting their faces to the camera and also using a registered user ID and password. They can communicate about their health situation through voice, chat, and video conversations securely with the robot service to request medical attention. Moreover, Triage-Bot automatically captures vital signs from facial video. Voice data is converted to text to extract a concise summary of the patient's health condition, while emotions are detected from both text and video data. The processed patient information is linked to the patients' Electronic Medical Records to automatically classify the patients into one of the five severity level as defined by the Canadian Triage and Acuity Scale. Currently, we are working on the evaluation of the services by medical professionals.**

***Index Terms*—TriageBot, Canadian Triage and Acuity Scale, Automatic Speech Recognition, Electronic Medical Records, photoplethysmography**

I. Introduction

COVID-19 emphasized the critical need to equip the hospital Emergency Departments (ED) with on-demand resource management capability and digital technology to handle sudden and/or consistent influxes of patients with a spectrum of healthcare needs, spanning from severe life-threatening conditions to milder concerns. Studies have found that around 50% of ED visits are for non urgent reasons, incurring unnecessary costs and multiple adverse consequences [1]. Opting for a visit to the ED instead of seeking care at alternative sites such as a physician's office, retail clinic, or an urgent care facility for non-urgent conditions can result in excessive healthcare expenses and unnecessary diagnostic tests. A recent study projected $4.4 billion in annual savings if non-urgent ED visits were cared for in retail clinics or urgent care centers during the hours these facilities are open [3] [4].

Every individual seeking emergency medical attention must undergo an assessment of criticality categorization according to the acuity of their health conditions [2]. Based on this assessment, patients are prioritized for treatment. This categorization procedure is commonly known as triage. The primary objective of triage is to efficiently allocate ED resources and optimize their utilization across the patient population. Numerous triage instruments are in use around the world, including the Canadian Triage and Acuity Scale (CTAS), the Manchester Triage System, the Australasian Triage Scale, and the Emergency Severity Index [2]. CTAS employs a five-tiered approach to categorize patients within an ED, streamlines patients' flows through the hospital, and equitably distributes the workload. It relies on an extensive array of patients' complaints as the basis for determining their triage level. The widespread adoption and integration of CTAS into Canadian healthcare standards make it an ideal choice for our research and analysis.

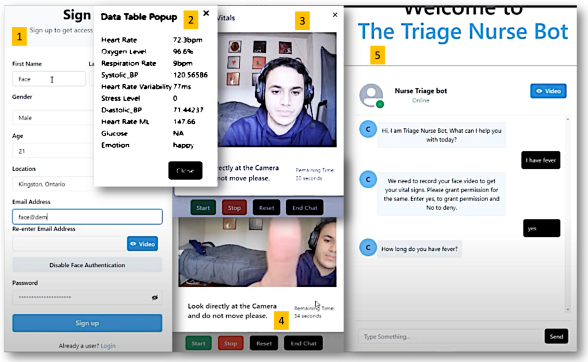


Fig. 1. From left to right: (1) Sign-up page for user authentication; (2) Pop-up table showcasing vital signs of the patient; (3) and (4) Video screens for capturing patient's face or finger videos; (5) TriageBot with patient conversation

Considering the landscape of innovative medical chatbot systems, New York-based One Remition distinguishes itself with a user-friendly interface and personalized mental health support, harnessing Artificial Intelligence (AI) for interactive conversations. Youper specializes in enhancing emotional well-being and mental health through conversational therapy techniques and mood tracking, offering users personalized insights and coping mechanisms. In 2013, Babylon Health vowed to enhance accessibility and convenience by employing AI to analyze user-input symptoms and offer medical advice, yet faced historical challenges due to inaccuracies [6]. Positioned as a "personal nurse," Florence concentrates on aiding individuals with interactive features, including reminders and

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health tracking. Ada Health simplifies healthcare journeys with 24/7 personalized health assessments and reduces unnecessary visits; however, its effectiveness relies on the accuracy of user input [7]. Sensely serves as a virtual medical assistant, engaging users in conversation while gathering information to assist in decision-making and determine patients' urgency [8]. Buoy Health guides users through personalized symptom-checking experiences, asking pertinent medical questions to offer insights.

Existing systems lack dual support for video and voice, remote vital sign monitoring, emotion detection, robust privacy, secure video streaming, and integration with Electronic Medical Records (EMRs). We propose Triage-Bot, an online triage system, to address the above limitations and predict the type and severity of illnesses. It can reduce ED visits and waiting times leading to substantial cost savings. Triage-Bot collects information about patients' health condition, categorizes the severity of the condition, and presents the assessments with the collected information to the human triage nurse or health practitioner in service to provide a response to send back to the user. The novelty of the proposed system lies in the following capabilities.

1) *Security*: Allows new users to register and old users to login [12] to a secure service platform and perform secure multiparty computation [11] to ensure data security.
2) *Multimodal User Interaction*: Provides accessible online user interface for multimodal audio, video, and chat conversations and other user interactions [14].
3) *Vital Sign Measurement*: Vital signs are measured with users' approvals from face or finger videos [5] [13].
4) *Emotion Recognition*: Emotions are detected from video and text data to enable empathy and compassion in conversation and to improve criticality assessment.
5) *Dialogue Generation*: Generate goal-oriented dialogue to collect necessary information for disease diagnosis and criticality assessment [10] [14].
6) *Consult EMR*: Extract patients' historical health data from an EMR system [9].
7) *Disease Prediction*: Predict a diagnosis based on the acquired information [9].
8) *Severity Assessment*: Categorize patients' severity using the CTAS scale based on collected data.

A large spectrum of health situations may be presented to a real system. In this pilot study, we have limited the possible conditions to COVID, chest pain, and mental health conditions for exploration and validation purposes. The dialogue generation, disease prediction, and patient ranking modules consider only these three possible disease cases in this pilot study to explore the challenges, feasibility, and potential of expanding the system to support a wide spectrum of health situations. The objective of this research is to leverage the healthcare system with a tool that can support the triage system so that patients can get an initial assessment online to reduce ED visits for non-urgent health conditions. Some of the screenshots from the system are compiled and presented in Fig. 1.

II. Methodology

A. Design Objectives and Architecture

Based on study of the state-of-the-art (SOTA) solutions and consultation with nurse practitioners and researchers in health sciences, we identified the critical features needed in a Triage-Bot system. We followed modular architecture where each of the functional features as listed in Section I, is implemented as a separate module and later integrated to construct the robot. As such, the work involved efforts of many researchers and developers. We currently have most of the modules implemented with the final evaluation pending, which requires recruitment of study participants.

B. Challenges

Some of the key design and implementation challenges identified in this study include: a) handling inaccurate, exaggerated, incomplete, or deceiving information presented by the patients which a human nurse can sense, b) handling sensitive information such as user data including videos and enabling secure query into EMR and data processing, c) generating dialogues effectively to collect necessary information with minimum but compassionate interactions, d) guiding user through the vital sign measurement process, and e) compiling all the data to determine possible high level diagnosis to categorize the severity of the patient.

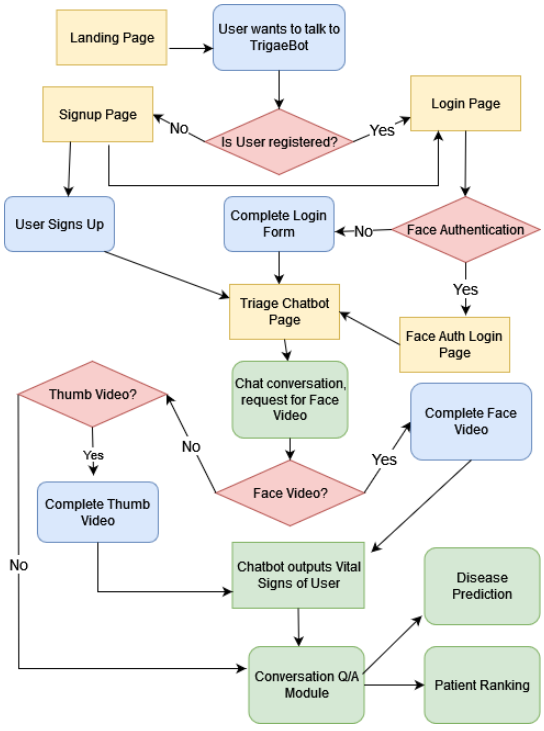


Fig. 2. Triage-Bot Procedure and information flow

C. Feasibility and Evaluation

To understand the complexity and feasibility of developing such a system and evaluate the functionality and usability,

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we limited the possible diseases at a high level to COVID, chest pain, and mental illness as these types of illness are handled differently by the ED triage nurses. Detecting the type of problem will allow Triage-Bot to forward the collected information with a tag to the proper care unit. We also explored the ethical concerns from the perspective of responsible AI including the explainability of the diagnosis and decision modules which led to interesting future research ideas. It is critical for the systems to be thoroughly evaluated by real users for practical applications. With the support from the government and industry, it is feasible to develop and deploy a Triage-Bot system initially at the ED, healthcare clinics, or community care facilities.

D. Implementation

We applied a distributed development strategy by adopting a modular design approach and a bottom-up strategy to design each module. All our modules have been implemented and integrated to demonstrate a cloud-based, mobile-accessible Triage-Bot, accessible anywhere. Due to space constraints, in this work-in-progress paper, we explain each module below at a high level and present some screenshots. The complete workflow of the Triage-Bot is given in Fig. 2.

Backbone: We built a web-based infrastructure [13] with a core MongoDB database to support the functionality and integrate all the modules. A web-server allows user to access the back end application modules.

Functional Modules: Module-1 enforces security measures with both password based login and biometric-enabled face authentication [12]. We also implemented a secure multiparty computation system to allow the system and the healthcare workers to query and analyze data if needed [11]. Module-2, multimodal user interaction, enables real time communication with automatic speech recognition, voice transcription, chat, and video capturing capabilities to allow hands-free accessible interaction (see Fig. 1) [14]. Next.js, React.js, Chakra UI, and javascript are used to ensure access through mobile devices and real time performance. Module-3, vital sign measurement, applies rPPG (remote Photoplethysmography) technology to measure heart rate, heart rate variability, oxygen saturation, blood pressure, and blood glucose from face and finger videos (see Fig.1) [5]. A variety of techniques are used to clean the captured data, extract the Blood Volume Pulse signal, and process it using different deep learning and statistical methods to estimate the required vital signs. Module-4, recognizes emotion [15] from video and text data using different deep learning models while computing vital signs to influence dialogue generation and disease diagnosis. Module-5 generates dialogues based on pre-defined decision trees created in consultation with nurse and medical practitioners for the 3 selected disease cases to learn about users' health conditions. In the future, we plan to extend this module using trained ML and generative Large Language Models. Module-6 focuses on linking with EMR which is simulated as a data table populated with MIMIC III benchmark data [11] for patient ranking. Module-7, implements a rule-based algorithm in this pilot study to diagnose the disease based on (i) user reports for the 3 selected disease cases, (ii) vital signs measured by Triage-Bot, and (iii) the EMR data. We plan to extend it in the future to incorporate trained ML models to diagnose diseases and report uncertainty [9]. Module-8 implements the CTAS algorithm to assess the severity of the patient's health condition on a scale of 1 to 5 (1: extremely severe, and 5: mild). The assessment is sent to the human administrator with a summary [11] of the collected data who reviews the information and returns a final response for the user.

III. Conclusion

This pilot study was designed to explore the potential, challenges, risks, and feasibility of a Triage-Bot system. Feasibility study has been completed and preliminary feedback has been obtained from collaborating health practitioners. Further work is needed to recruit participants and validate the various modules separately to improve accuracy. With proper validation and explainability through the rules, such systems can make a great impact on the ED triage process.

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